

# AKASH SURIYA NARAYANAN

Via Monte Cengio 19, Padova, Italy 35138 • ak9962851601@gmail.com • +39 3514184915 • +91 9962851601

LinkedIn: [linkedin.com/in/akash-chemical](https://www.linkedin.com/in/akash-chemical) • Portfolio: [akashsn.com](https://akashsn.com)

## PROFESSIONAL SUMMARY

Chemical engineer with 2+ years of industry experience at a global abrasives manufacturer, currently pursuing an M.Sc. at the University of Padova. Hands-on expertise in Aspen Plus process simulation, CAPEX project execution, and production optimisation. Experienced in applying engineering principles to real industrial challenges, with a strong focus on sustainable and efficient process design. Erasmus+ alumnus with multicultural project experience across five European universities.

## EDUCATION

**M.Sc. Chemical and Process Engineering** | University of Padova, Italy

Sep 2024 – Dec 2026 (Expected)

*Industrial Thesis: Aspen Plus process simulation project (company details confidential)*

*Relevant coursework: Multiphase Thermodynamics, Separation Unit Operations, Process Design, Process Fluid Dynamics and Simulation, Risk and Safety Analysis, Machine Learning for Process Engineers*

**B.Tech Chemical Engineering** | SASTRA Deemed University, India

Aug 2022

CGPA: 8.011/10.00 | 85th Percentile

## WORK EXPERIENCE

**Executive – Production** | Carborundum Universal Limited (CUMI), India

Aug 2022 – Aug 2024

- Led production of hot-pressed resinoid grinding wheels, achieving a 75% increase in output through process optimisation and TPM-driven improvement initiatives.
- Secured multimillion-rupee CAPEX approvals by preparing comprehensive technical reports and business cases for the Senior Leadership Team.
- Commissioned multiple CAPEX projects in coordination with cross-functional teams and external vendors, delivering on time and within budget.
- Built automated Power BI dashboards for 3 manufacturing units, improving production planning visibility and operational decision-making.
- Led product portfolio expansion of hot-pressed wheels using Quality Function Deployment (QFD) methodology.

## INTERNATIONAL PROGRAMS

**Erasmus+ Blended Intensive Programme (BIP)** | Hochschule Kaiserslautern, Germany

May 2025

*Decarbonisation Strategies in Process Engineering*

- Completed online collaboration and 1-week physical mobility with students from University of Padova, University of Valladolid, UPC Barcelona, University of Turku, and Hochschule Kaiserslautern.
- Designed and simulated an MVR distillation column for ethanol purification using Aspen Plus; presented simulation results and attended a Sulzer technical lecture.
- Visited BASF Ludwigshafen Verbund site and Berkel AHK Alkoholhandel GmbH for industrial insights into sustainable chemical processes.

## PROJECTS

**Process Plant Conceptual Design and Simulation** | University of Padova – M.Sc. Process Design Project

Sep 2025 – Feb 2026

- Collaborated in a team of 6 to conceptually design a 5,000 t/year speciality chemical production plant using Aspen Plus, following a hierarchical design approach covering economic potential screening, CSTR simulation, and sensitivity analysis to optimise reactor conversion and recycle purge composition.
- Designed a multi-column distillation separation system achieving 99.9% molar product purity; completed full cash flow and profitability analysis confirming economic viability.

**Process Optimisation via VSM Methodology** | Carborundum Universal Limited

Aug 2022 – May 2023

- Calculated OEE and Manufacturing Cycle Efficiency (MCE) across all production processes using Value Stream Mapping; implemented targeted kaizens that increased hot-pressed grinding wheel output by 75%.

**Power BI Dashboard for Multi-site Operations** | Carborundum Universal Limited

Sep 2023 – May 2024

- Built an automated, interactive Power BI dashboard using Excel macros to consolidate ERP data across 3 manufacturing locations, enabling SKU-level operational insight and faster decision-making.

### Model Development for Boiler Efficiency | CORI Rubbers Pvt. Ltd.

Mar – Jul 2022

- Developed an Excel model for boiler efficiency determination with sensitivity and error analysis; performed HAZOP analysis and authored SOP for the softener plant.

### CFD Study – Flow Through Single-Holed Orifice in Series | SASTRA University

Aug – Dec 2021

- Numerically estimated pressure drops and cavitation potential using ANSYS Fluent; validated against experimental data and compared steady-state vs transient behaviour.

## TECHNICAL SKILLS

---

**Expert:** Aspen Plus, Power BI, Excel (Advanced), Microsoft Office Suite

**Proficient:** ANSYS Fluent, AutoCAD, Pro/II, MATLAB, Minitab, Excel VBA

**Familiar:** C, C++, SolidWorks, DWSIM, Scilab

**Process Methods:** HAZOP, Six Sigma, Design of Experiments, QFD, TPM, VSM, Sensitivity Analysis, Cash Flow and Profitability Analysis

## CONFERENCES AND WORKSHOPS

---

- Presented paper on CFD simulation at SCHEMCON 2021, Indian Institute of Chemical Engineers – Sep 2021
- Workshop on Quality Function Deployment, Lead Training Solutions – Jun 2023
- Workshop on Design of Experiments (DOE), Confederation of Indian Industry (CII) – Apr 2023
- Workshop on Value Stream Mapping, Confederation of Indian Industry (CII) – Dec 2022

## CERTIFICATIONS

---

- Lean Six Sigma Yellow Belt (ISYB), Anexas Europe – Jul 2020
- Oil and Gas Industry Operations and Markets, Duke University, Coursera – Aug 2020
- Leadership and Emotional Intelligence, Indian School of Business, Coursera – Sep 2020
- Employment Communication, IIT Kharagpur, NPTEL – Mar 2021

## AWARDS AND RECOGNITION

---

- |                                                                     |          |
|---------------------------------------------------------------------|----------|
| • Star of the Quarter – FY24, Carborundum Universal Limited         | May 2024 |
| • First Prize – Industrivia, Siphon 7.0, SVNIT Surat                | Mar 2022 |
| • First Prize – Shipwreck, STEER, SSN Chennai                       | Mar 2022 |
| • Second Prize – Paper Presentation, ChemIgnite, Manipal University | Nov 2021 |

## LEADERSHIP

---

- |                                                                           |                     |
|---------------------------------------------------------------------------|---------------------|
| • Chairperson, IChE Students' Chapter, SASTRA University                  | Aug 2021 – Aug 2022 |
| • Designer, National Conference on Advances in Process Engineering (CAPE) | Jan – Mar 2021      |

## LANGUAGES

---

**English, Tamil** (Read, Write, Speak) **Hindi** (Read, Write)